UIST 2026 · POSTERS

BrowserLens

Interactive Diagnosis of Web-Agent Trajectories

with a Layered State–Action Graph

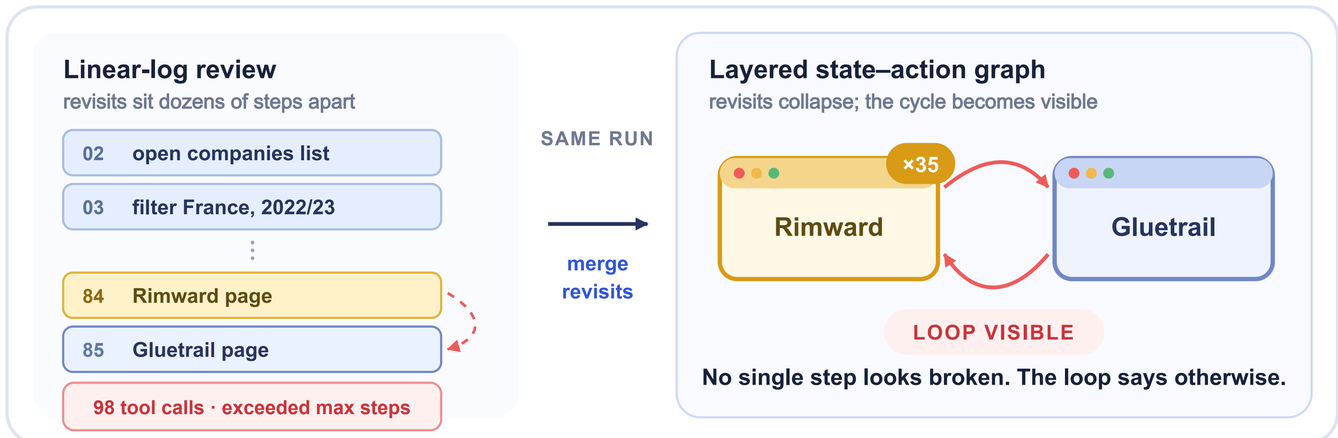
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1 THE REVIEW PROBLEM

A completed run leaves only a trajectory. Its linear log hides cross-step structure.

Revisits, loops, dead ends, and stalls are scattered across distant entries and reassembled by the reviewer.

Design move organize by browser-state identity



2 SEE THE STRUCTURE, THEN DRILL DOWN

One trajectory. Three graph layers.

Browser state is the unit of diagnosis

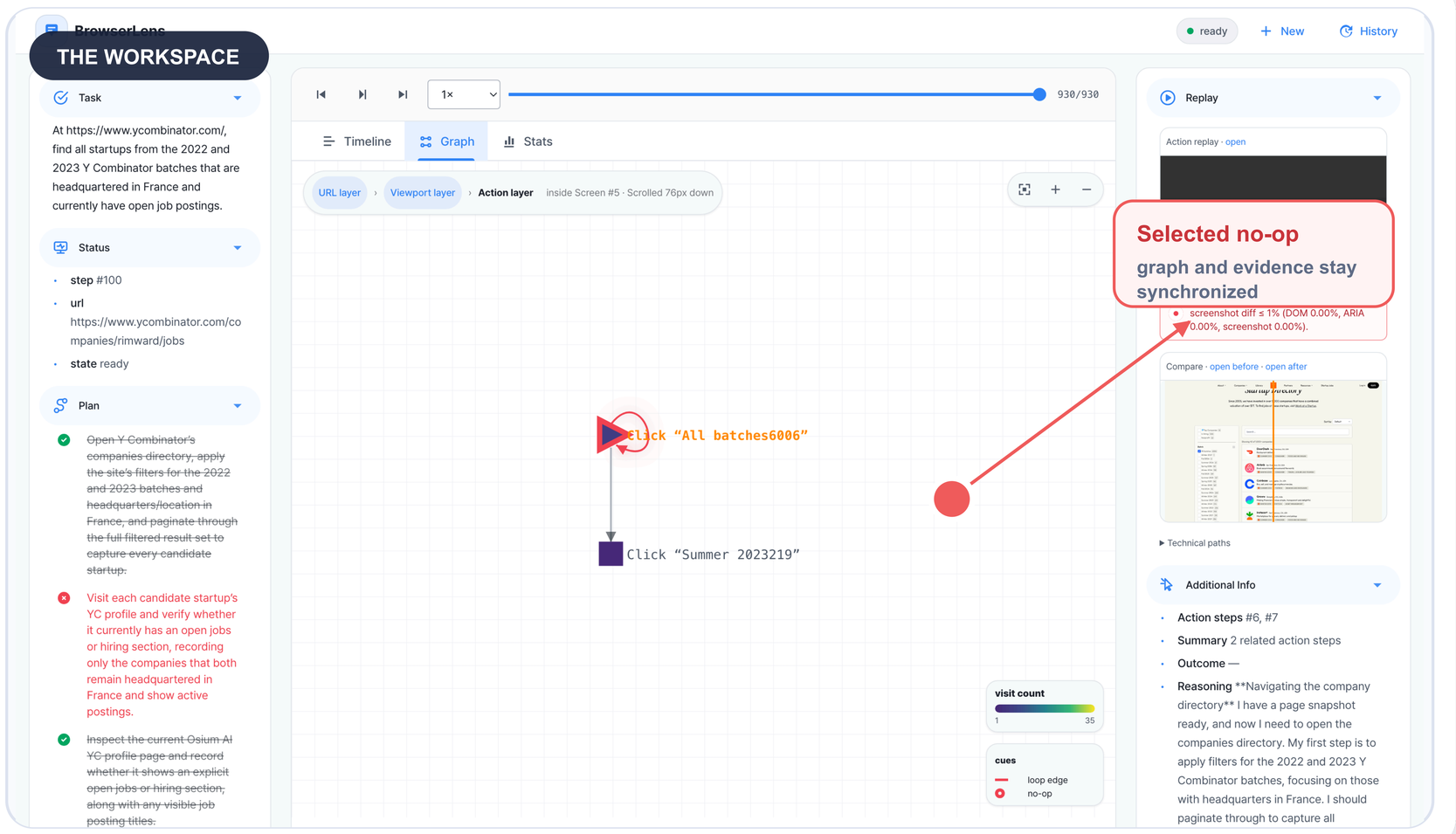
BrowserLens merges logged instances that share a deterministic identity.

1 **URL layer** — which page?

2 **Viewport layer** — which screen?

3 **State-Anchored Action layer** — what was done there?

Read-only diagnosis
for completed, logging-enabled runs.
The graph indexes logged states and transitions.

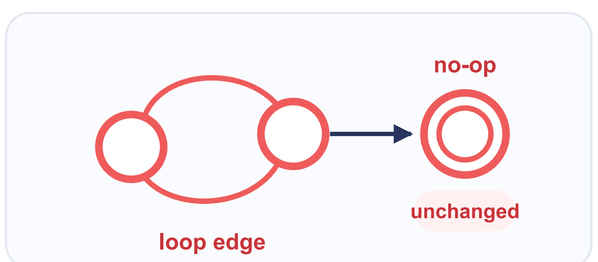


3 REVIEW FOLLOWS THREE STAGES

Visual cue → tentative failure points → evidence

A cue never asserts a failure; recorded evidence decides.

1 Visual cue



Repeated-node highlight · Loop-edge highlight · No-op highlight · Hidden-cue badge

Four visual cues mark suspicious structure.

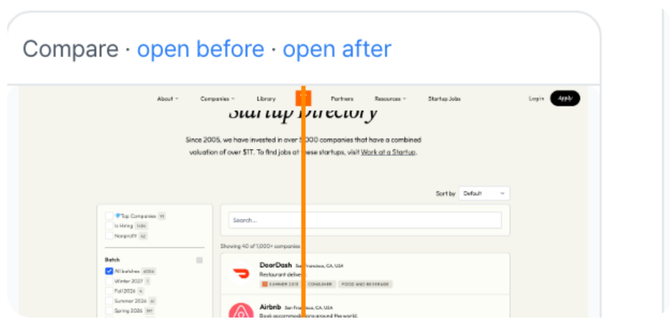
2 Tentative failure points

Cues and failure hypotheses relate many-to-many; the reviewer verifies them against recorded evidence.

repeated state · repeated action · navigation loop · dead-end · navigation stuck · grounding error

N : M relationship — candidates, not verdicts

3 Evidence decides



Before/after · DOM/ARIA diff · action/tool log
Replay · model context
The evidence confirms the click changed nothing.

4 TAKEAWAY

Organize the run around browser state, not chronological time.

BrowserLens makes cross-step structure visible.

SCOPE

Read-only by design

Completed, logging-enabled runs.
Steering and repair remain future work.

Identity may split volatile URLs or merge distinct screens.

NEXT

Qualitative comparison

Within-subjects study against linear-log review on failure-rich runs: how do reviewers find and explain flawed trajectories?

No speed or accuracy claims